What Teacher-Forced Loss Misses: Stopping at the Goal in Fine-Tuned Vision–Language Navigation Policies

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Abstract

Vision-and-language navigation agents follow a written route instruction through a building using only their own camera images, and an episode counts as a success only if the agent itself stops near the goal. Running every saved checkpoint in a simulator is expensive, so a general-purpose fine-tuning pipeline can instead keep the checkpoint with the lowest validation loss. For a navigation policy this loss is teacher-forced: the model sees the expert's past frames and is scored on whether it predicts the expert's next action. We fine-tune the Qwen3-VL-8B vision–language model on Room-to-Room in Continuous Environments (R2R-CE), with and without class weighting, and run the checkpoints this loss selects in the Habitat simulator, in buildings seen during training. None of the selected checkpoints reaches the goal and stops there. Those trained without class weighting never stop and behave like a baseline that always moves forward, and class weighting changes which action the policy repeats. A preregistered follow-up executes every saved checkpoint, and no checkpoint was shown to navigate better than the selected one. The validation loss gives no warning, because stopping is rare among the decisions it averages, as earlier work has noted. Two offline checks flag every run before any simulator test: recall on the STOP action, and a pass/fail test fixed before the first run. In this setting, a low validation loss is not evidence that a policy can navigate. Code and run records are at `github.com/vivekmattam02/Stop_Blind`.

1 Introduction

Vision-and-language navigation (VLN) asks an agent to follow a written route instruction through a building, seeing only what its own camera sees. In Room-to-Room in Continuous Environments (R2R-CE) [Anderson et al., 2018, Krantz et al., 2020], the agent has four actions at every step: move forward 0.25 m, turn left or right by 15 degrees, or stop. An episode counts as a success only if the agent itself stops within 3 m of the goal. Stopping is a spatial judgment: the agent must recognize from its own view that it has reached a place described only in words. We report Success Rate (SR), the share of episodes that end this way, and Oracle Success (OSR), the share whose path ever enters the goal radius, whether or not the agent stops there.

Fine-tuning a navigation policy saves a series of checkpoints, and one of them must be chosen. The direct way to choose is a closed-loop rollout: run each checkpoint in the simulator, where every action it takes determines what it sees next, and count its successes. Rollouts are expensive, but without them a general-purpose fine-tuning pipeline can still compute one number per checkpoint, its validation loss. Asked to keep its best checkpoint, the Hugging Face Trainer keeps the one with the lowest evaluation loss unless given another metric [Hugging Face, 2026]. For a navigation policy

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this loss is teacher-forced. At each held-out step the model sees the expert's own past frames, and it is scored by the cross-entropy (CE) of the expert's next action.

In R2R-CE, this loss averages over decisions in which one action dominates. Of the 631,244 decisions in the released training split, 64.1% are MOVE_FORWARD. Among the 197,455 decisions eligible for this study, 66.6% are MOVE_FORWARD and only 1.76% are STOP. The average can therefore improve while the model stops predicting STOP, the action that success requires. Figure 1 shows a case from our experiments. The checkpoint the loss selects is scored on 256 held-out decisions, only three of which call for STOP. In 61 episodes in the simulator, it never stops. We call this failure mode STOPBLIND: when STOP is only a few percent of the decisions that a selection loss averages, the loss can rank best a checkpoint that never stops at the goal, and it gives no sign of whether any checkpoint does.

Xiang et al. [2020] pointed out that an unweighted loss gives STOP too little attention. Wijmans et al. [2019] warned that navigation policies can reach very high validation accuracy on action prediction while missing every change of action. Executing five low-loss checkpoints per model in embodied question answering, they found that a loss weighted toward those changes predicted end-to-end performance better than plain cross-entropy. Failures to stop have been documented for VLN agents on a discrete navigation graph [Zhao et al., 2023] and reported for a vision–language model (VLM) fine-tuned for R2R-CE [Dang et al., 2026]. To our knowledge, no one has measured which checkpoint the loss selects for a VLM policy in continuous R2R-CE, or whether it passes over a better one.

We fine-tune Qwen3-VL-8B, an open VLM, with and without class weighting, three seeds each. Class weighting multiplies each decision's loss by the inverse of its action's training frequency, so that rare actions count more. We run every selected checkpoint in the simulator on a frozen set of 61 episodes from `val_seen`, the validation split whose buildings also appear in training, under the official 500-step cap. We compare each run with an always-forward baseline, a policy that moves forward at every step. Before the first run, we also fixed a pass/fail test, the existence gate, which asks whether a model learned more than how often each action occurs and whether it recalls STOP (Section 3). In a preregistered follow-up, we then execute all 37 distinct saved checkpoints to test whether the rule passed over a better one. Its analysis and the bar a checkpoint must clear were fixed before any unselected checkpoint was run or scored. The study covers one model, one training budget of 192 updates, and buildings seen in training. Three single-seed probes, which supervise four-action chunks, scale the data $44\times$, or add metric 3D geometry, change the policies' behavior but cannot test stopping, because their data loader dropped every decision whose target was STOP. Our contributions are:

1. **Without class weighting, the validation loss selects policies that never stop.** In all three unweighted runs, the loss selects the same checkpoint, update 160 of 192. It predicts STOP on none of the 256 held-out decisions used to select it, and on none of the 64 STOP decisions in a separate held-out set. In the simulator, two of these checkpoints take the same actions as the always-forward baseline, and the third differs in one episode.
2. **Class weighting changes which action is repeated, and the policies still fail.** Weighted policies use all four actions in aggregate, but most episodes repeat one action throughout, and every STOP they press is outside the goal radius.
3. **No selected checkpoint succeeds, and no other was shown to do better.** All six selected checkpoints have SR $0/61$. Of all 37 distinct saved checkpoints, one succeeds in one episode: the untrained model, $1/61$. No candidate reaches the preregistered bar against its run's selected checkpoint.
4. **Per-class checks detect the failure offline.** Recall on STOP and the existence gate flag all six selected runs before any simulator run, and none of the 37 checkpoints passes the gate.

2 Related work

Rare actions in navigation imitation losses. The closest prior work concerns rare decisions under an unweighted imitation loss. In embodied question answering, Wijmans et al. [2019] warned that policies that repeat the previous action "achieve very high validation accuracy for action prediction but miss every transition between actions". The R2R-CE reference baseline, VLN-CE (VLN in

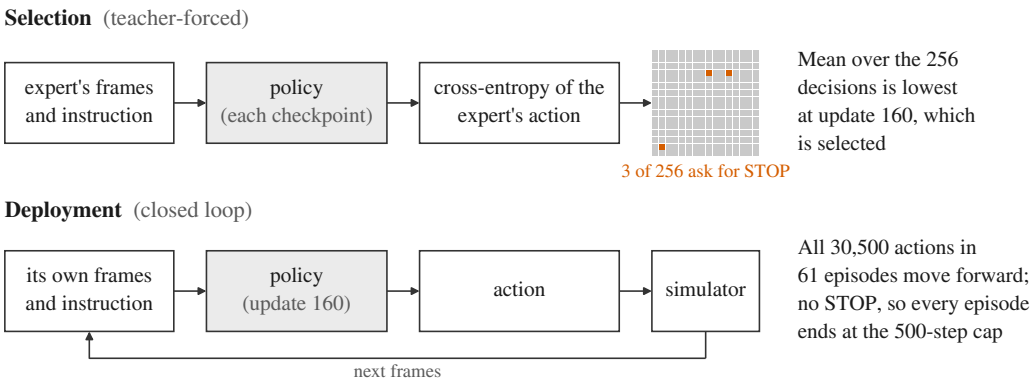


Figure 1: **Checkpoint selection compared with deployment.** Selection scores each of 256 held-out decisions with the expert's history supplied, and picks update 160. Deployment runs that checkpoint on 61 episodes under the 500-step cap, where its own action sets the next observation. All 30,500 actions it takes are MOVE_FORWARD, although three of the 256 scored decisions ask for STOP. Shown for unweighted seed 20260826. The weighted runs behave differently (Table 2).

Continuous Environments), trains with their inflection weighting, which upweights steps where the expert's action changes [Krantz et al., 2020]. Xiang et al. [2020] argued that STOP, which occurs once per episode, "will receive less attention if we treat all actions equally", and added a dedicated stopping head. NaVILA, a VLM fine-tuned for R2R-CE, rebalances its labels, in particular because STOP is underrepresented. Removing the rebalancing lowers success on `val_unseen`, the split of unseen buildings that this study never opened, from 49.7 to 30.0 [Cheng et al., 2025]. A small VLM fine-tuned for aerial navigation was "critically affected by a severe action imbalance in the training data" [Tamazyan et al., 2025]. On the discrete navigation graph, agents in four systems pass the goal without stopping [Zhao et al., 2023], and a case study of one agent finds that its errors reflect a forward bias learned in training [Yang et al., 2023]. These works change training or analyze trained agents. We instead measure which checkpoint plain cross-entropy selects for a VLM policy in continuous R2R-CE, where success depends on the rare STOP action.

Offline metrics and closed-loop competence. In driving, offline prediction error need not correlate with driving quality [Codevilla et al., 2018], and open-loop metrics on a modern planning benchmark favor planners that nearly ignore their input [Dauner et al., 2023]. In manipulation, checkpoints selected on validation data are 10–100% worse than those selected by rollout [Mandlekar et al., 2021], and choosing a trained policy's input-modality mask by lowest validation loss picked a mask with zero success in all three seeds [Hao et al., 2023]. Behavior-cloned policies can reach near-zero held-out error by relying on spurious cues that fail at deployment [de Haan et al., 2019], including collapse toward a dominant action [Codevilla et al., 2019]. Two results narrow the gap: better-chosen validation data improve the correlation between offline and online scores [Codevilla et al., 2018], and under log-loss with a discrete action vocabulary, compounding error grows at most polynomially with the horizon [Foster et al., 2024]. Our setting differs from these studies in two ways: success in navigation hinges on one rare action, and the offline metric we test is plain cross-entropy used to select checkpoints.

Checkpoint selection in R2R-CE. R2R-CE papers that state a selection rule select on closed-loop `val_unseen` performance [Krantz et al., 2020]. NaVILA trains for one epoch and states no selection rule [Cheng et al., 2025]. We found no R2R-CE paper that selects checkpoints by teacher-forced loss. We select by it deliberately, execute every checkpoint it chose among, and leave `val_unseen` sealed. Appendix B discusses this work in more detail.

3 Experimental setup

Data. We train on the released R2R-CE training split [Krantz et al., 2020]: 197,455 eligible decisions from 3,603 physical trajectories in 61 scenes. For testing in the simulator, we froze a separate

set of 61 `val_seen` episodes spanning 53 scenes. These episodes were chosen without reference to any policy's results. They are the hardest expert path in each scene (the most blocked forward steps, then the most actions and turns), plus every other trajectory whose released shortest-path length differs from the simulator's, 12 such trajectories in all. None of them is used in training or in any held-out set. We split the training data by physical trajectory, so the several instructions written for one path always fall in the same part. Of the 3,603 eligible trajectories, 2,451 supply 3,072 training decisions. The rest supply one held-out decision each: 256 for checkpoint selection (the selection set), 640 for the natural set, whose actions keep their usual proportions, and 256 for the balanced set, with 64 per action. Scenes are Matterport3D scans [Chang et al., 2017], used under its terms of use [Matterport, Inc., 2017] and rendered in Habitat-Sim.

Model and training. For each decision, the model receives the instruction and the last four camera frames, and predicts one of four action tokens: STOP, MOVE_FORWARD, TURN_LEFT or TURN_RIGHT. A decision is step t of an expert trajectory, with $3 \leq t \leq 96$, and its frames are the RGB images at steps $t-3$ to t (224×224 pixels, 90-degree field of view), rendered by replaying the expert's actions. The policy receives no depth. Training and scoring are therefore teacher-forced, with every frame taken from the expert's path. The loss is the cross-entropy over the four action tokens (Appendix D), and the predicted action is the highest-scoring one, with an exact tie going to the lowest action ID. In the simulator, the frames are the policy's own, and the first three steps are padded with copies of the first frame. We freeze Qwen3-VL-8B-Instruct [Bai et al., 2025] (Hugging Face revision `0c351dd`) and train only a rank-32 adapter (LoRA) [Hu et al., 2022] on its attention layers. We train with AdamW (learning rate 2×10^{-5}, weight decay 0.01, gradient clipping at 1.0, microbatch 1, gradient accumulation 16) for 192 updates. The only difference between the two treatments is inverse-frequency weighting, normalized so that the mean sample weight is one. The three seeds, 20260825, 20260826, and 20260827, change only the adapter's initialization and the order of examples. Everything else is identical across runs, including the data, the held-out sets, the model, the optimizer, the selection rule and the thresholds.

Checkpoint selection and the existence gate. Each run saves a checkpoint every 32 updates, including the untrained model at update 0. The selection rule keeps the checkpoint with the lowest mean cross-entropy on the selection set (the selection loss), and the earlier one on an exact tie. The existence gate, fixed before the first run, asks whether a model has learned more than how often each action occurs, by comparing it with two baselines that ignore the input. On the natural set, a model is compared with the frequency-only baseline, which always predicts the training action frequencies and whose loss is the entropy of those frequencies, $H = 0.9247$ nats (natural-log units). On the balanced set, it is compared with a uniform guess, whose loss is $\ln 4 = 1.3863$ and whose accuracy is 0.25. The existence gate passes only if three conditions hold. The natural-set loss must fall below $H - 0.05$ and the balanced-set loss below $\ln 4 - 0.05$, both judged by the upper end of a 95% interval from resampling scenes, and balanced STOP recall must be at least 0.25.

Simulator evaluation. Our pinned Habitat-Sim evaluator [Savva et al., 2019] reproduces all 778 released `val_seen` expert trajectories and their official metrics. Besides SR and OSR, we report SPL, success weighted by path length, and SDTW, success weighted by how closely the path follows the expert's (normalized dynamic time warping, nDTW) [Ilharco et al., 2019]. Before any of the six training runs, we fixed a suite of 14 runs of non-learning policies on the same 61 episodes, 854 episode-runs in all (Table 3). Replaying the expert's trajectory succeeds on $61/61$ episodes with SPL 0.9679, while random and handcrafted policies each succeed on 3 of their 305 episodes, so the evaluator does not force zero success. Learned policies run on the same 61 `val_seen` episodes (53 scenes) as the controls. Habitat executes 0.25 m forward moves and 15-degree turns, with a 3 m goal radius and the official 500-step cap. Comparisons are paired by seed and episode. Intervals come from 10,000 fixed-seed resamples of the 53 scenes, keeping all episodes and seeds of each sampled scene.

Executing every saved checkpoint. We executed all 37 distinct saved checkpoints on the same 61 episodes, all on one GPU type (NVIDIA L40S): the 30 the rule passed over, the untrained model, and the six selected ones again for comparison. Each passed-over checkpoint is compared with its run's selected checkpoint, and the untrained model with all six. A candidate counts as better only if it passes a paired test fixed in advance (a one-sided exact McNemar test with Tarone's correction for discrete tests [Tarone, 1990]) and then passes it again on 61 new episodes (Appendix F).

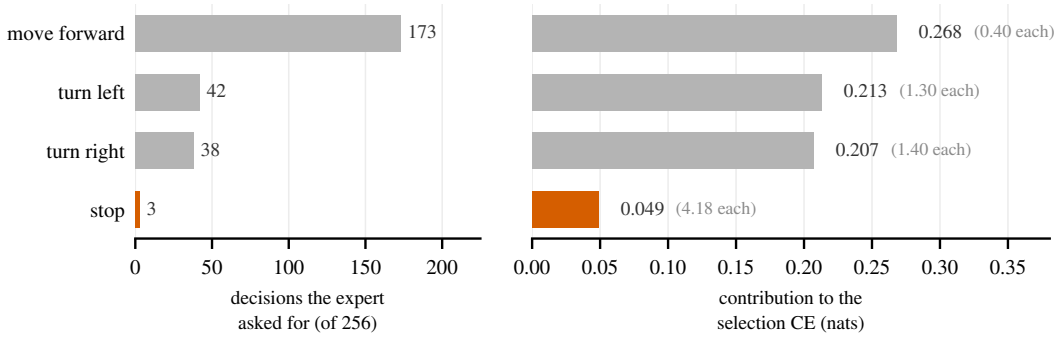


Figure 2: **Decomposition of the selection loss.** It is a mean over 256 decisions, so each expert action contributes its own cross-entropy weighted by how often the expert asks for it. The four contributions sum to the reported 0.7373 with no residual. STOP has the worst per-decision cross-entropy, 4.18 against 0.40 for MOVE_FORWARD, and this checkpoint never predicts it. At three decisions in 256, that failure moves 6.6% of the number. Update 160, unweighted seed 20260826; STOP's share is 3.5–6.2% across the other five runs.

Verification. Every input and output is recorded with its hash. Separate verifier programs reproduce the six training outcomes, all 366 learned rollouts, and the 61 matched always-forward runs without error. For the 37-checkpoint execution, every count was computed again from the raw episode records, independently of the preregistered analysis script. All tables and figures are generated from these records.

Preregistration and later additions. Only the first run, unweighted at seed 20260825, was fully specified before any result, in a written protocol whose SHA-256 hash the run records. We added the class-weighted runs after that run collapsed, the other two seeds after both seed-20260825 runs finished, and the simulator runs after all six. Each addition was fixed before it ran and kept the same selection rule and gate, and every seed is reported. The weighting comparison reuses the held-out sets that the first run had already used, so it is adaptive. The checkpoint-selection framing, the episode-level comparisons with the controls, and the scene-resampled intervals came after the results. Executing every saved checkpoint was added after review and preregistered on 9 October 2026, before any unselected checkpoint was executed or scored.

4 Results

The selection loss picks a checkpoint that never predicts STOP. In every unweighted seed, the run beats the frequency-only baseline on the natural set, and the rule selects update 160. Each selected checkpoint, however, predicts STOP on $0/256$ selection decisions and recalls $0/64$ of the balanced set's STOP decisions. Figure 3 shows the three selection decisions at which the expert stops. On the balanced set, each unweighted run's loss is worse than a uniform guess ($\ln 4$), although its accuracy is better than chance (0.25). Class weighting improves the balanced-set loss, accuracy, and STOP recall in $3/3$ pairs, but worsens the natural-set loss in $3/3$. Weighted STOP recall is $12/64$, $1/64$, and $6/64$, so the direction of the effect replicates across seeds but its size does not. All six existence gates fail (Table 1). Figure 4 plots the selection loss and the number of predicted STOPs over training. Figure 2 breaks one selected checkpoint's loss of 0.7373 into its parts. Exact score ties go to the lowest action ID, which is STOP (Appendix C).

Success is zero for every selected checkpoint. Table 2 summarizes the simulator runs, and Appendix E plots them. All six checkpoints obtain SR $0/61$, SPL 0, and SDTW 0. Each $0/61$ is still consistent with a true success rate of up to 5.9% (exact two-sided 95% Clopper–Pearson interval, treating episodes as independent). Each also has OSR $1/61$, but from different episodes. Every unweighted checkpoint enters the goal radius only on episode 562, and every weighted checkpoint only on episode 394. Episode 562 ends inside the radius, 2.356 m from the goal, after 480 decisions inside the radius without STOP. Episode 394 comes within 1.241 m of the goal, leaves the radius, and ends 4.513 m away after 33 decisions inside it without STOP. On these paths, a perfectly timed

Table 1: Three-seed paired replication. Values are mean $\pm$ sample standard deviation; deltas are weighted minus unweighted. All seeds are reported. Natural CE is scored on the 640-decision natural set; selection CE, on the 256-decision selection set that picks each checkpoint (Figures 2 and 4), is a different quantity.

Metric	Unweighted	Weighted	Paired delta
Natural mean CE ↓	0.754 ± 0.012	0.955 ± 0.016	0.201 ± 0.019
Balanced mean CE ↓	1.725 ± 0.117	1.452 ± 0.132	-0.273 ± 0.223
Balanced accuracy ↑	0.332 ± 0.004	0.397 ± 0.038	0.065 ± 0.042
Balanced STOP recall ↑	0/64 in 3/3 seeds	12/64, 1/64, 6/64	0.099 ± 0.086
Existence gate	0/3 passed	0/3 passed	—

Table 2: Closed-loop behavior under the official step cap on the same 61 `val_seen` episodes. Counts are episodes per seed. Stopped: the episode ends with STOP. Constant: one action repeated for the whole episode. AF exact: the same action sequence as the always-forward baseline.

Treatment	Seed	SR	OSR (episode)	Stopped	Constant	AF exact
Unweighted	20260825	0/61	1/61 (562)	0/61	61/61	61/61
	20260826	0/61	1/61 (562)	0/61	61/61	61/61
	20260827	0/61	1/61 (562)	0/61	60/61	60/61
Weighted	20260825	0/61	1/61 (394)	5/61	47/61	13/61
	20260826	0/61	1/61 (394)	0/61	42/61	14/61
	20260827	0/61	1/61 (394)	2/61	55/61	20/61

All six checkpoints have SPL and SDTW equal to zero.

stop would have succeeded in at most 1 of 61 episodes. Most episodes fail because the policy never reaches the goal, and on the one path that does reach it, the policy does not stop.

No checkpoint was shown to navigate better than the selected one. All 2,257 episode runs of the 37 checkpoints completed (Table 4, Appendix F). The only checkpoint that succeeds is the untrained model, with $1/61$ (episode 358, two-sided 95% interval 0.04–8.8%). It also presses STOP 50 times in other episodes. Each of the 36 trained checkpoints scores $0/61$. Of the 213 STOP presses across all 37 checkpoints, one falls inside the goal radius. No candidate reaches the preregistered bar against its run's selected checkpoint. The sweep was designed to separate two explanations for the selected checkpoints' zero success. Either the loss passed over a checkpoint that navigates, or no checkpoint it ranked was shown to navigate better than its pick. It found no evidence for the first explanation, within the statistical power given in Section 6. Re-executed in this sweep, the six selected checkpoints again succeed in no episode, and their actions match the first run except at near-ties, where two actions scored almost equally (Appendix F).

Two preregistered predictions failed. We predicted that choosing each run's checkpoint by the lowest balanced-set loss, a rule named before any unselected checkpoint was scored, would beat the averaged loss. Neither this rule nor choosing each run's last checkpoint solves any episode. The balanced loss flags the failure without finding a better checkpoint. All 18 trained unweighted checkpoints predict STOP on none of the 256 selection decisions, so we predicted that the 15 unselected ones would never press it in the simulator. That prediction failed: four of them press STOP 23 times, all on the first step, 6.6–12.9 m from the goal.

Unweighted policies reproduce the always-forward baseline. Seeds 20260825 and 20260826 match the always-forward baseline's actions and every metric on all $61/61$ episodes. Seed 20260827 matches on $60/61$, and across its 30,500 decisions it emits 30,499 forward moves and one right turn. Every unweighted run lasts the full 500 steps. Five of seed 20260827's matching episodes begin with an exact tie between forward and a turn (Appendix C), so counting only strict preferences, 177 of the 183 unweighted episodes are always-forward.

Weighting changes which action is repeated. Weighted policies emit all four actions in aggregate, but most of that variety comes from differences between episodes. In $47/61$, $42/61$, and $55/61$



Figure 3: **The selected checkpoint at the goal.** The selection set has three STOP decisions, and all three are shown. The checkpoint is the one the selection rule picks, update 160 of unweighted seed 20260826. It emits MOVE_FORWARD at all three, with probability 0.60 to 0.73, giving STOP between 0.009 and 0.034. Images are Habitat-Sim renders of Matterport3D scans, and no physical robot was used. Cross-entropy stays low because these are three decisions in 256.

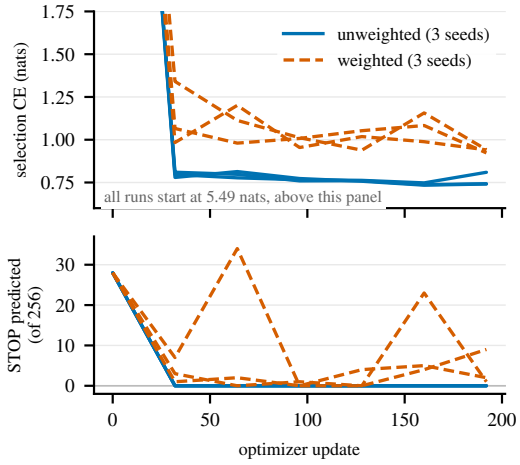


Figure 4: **The selection loss falls while the unweighted runs stop predicting STOP.** Cross-entropy on the frozen selection set, and how many of those 256 decisions the model calls STOP. Solid is unweighted, dashed weighted, three seeds each. Unweighted cross-entropy falls to about 0.8 within 32 updates and flattens while predicted STOP goes from 28 to zero and never returns. The weighted runs stay between 0.92 and 1.34, and their predicted STOP spikes to 34 and 23 at isolated points.

episodes, a single action is repeated from start to finish. Across each 61-episode suite, the policies switch actions only 30, 21, and 7 times, and the median longest run of one action is 500 steps in every seed. The three weighted checkpoints also produce identical action sequences on $28/61$ episodes.

Weighted policies stop only away from the goal. Of the 183 weighted episodes (three seeds $\times$ 61), seven end with STOP. All seven stops are outside the 3 m radius: six on step 1 and one on step 19, 6.63–12.80 m from the goal. During the only goal-region visits we observed, all six policies make 1,539 consecutive decisions inside the radius and never press STOP. These decisions come from only two episodes, so they are not independent trials.

5 Exploratory probes

We ran three single-seed probes, each aimed at another possible cause of the collapse: a training objective that rewards constant behavior, too little data, or a model without spatial grounding (Appendix A). All three train the model to predict a chunk of four actions at a time. Chunking makes a constant guess much less rewarding, since always giving the most common answer matches 64.1% of single decisions but only 25.2% of exact chunks. In the simulator, the chunk policy then uses all three movement actions, and constant-action episodes fall from $61/61$ to $39/61$. Two runs train on 40,000 and 134,754 decisions (the larger set contains the smaller) and reach best selection losses of 0.6070 and 0.5955; the chunk run above, trained longer on the original set, reaches 0.6843. Each is one seed, so we do not rank them. A third run adds a branch that feeds metric 3D geometry to the model, and the trained model uses it, since hiding those inputs raises its selection loss from 0.6407 to 1.0625. The branch reads the simulator's depth. On a robot it would take depth from a pipeline built for indoor time-of-flight failure, such as our self-calibrating fusion [Pushparaju et al., 2026].

These probes cannot test stopping. A chunk needs four future actions, so the data loader dropped every decision whose target was STOP, which is the last action of every expert trajectory. STOP then appeared in training only as the fourth action of a chunk, at 0.47–0.57% of supervised positions, and in only 4 of the selection set's 960 positions. No checkpoint that the two scaled runs and the geometry run scored during training predicted STOP, and in the simulator all three probe policies ran 30,500 actions each without pressing it.

6 Limitations

Development split only. All simulator results come from a frozen set of 61 `val_seen` episodes, a development split whose buildings appear in training. They support exact comparisons within this protocol. They do not support a leaderboard claim or a claim about unseen buildings (`val_unseen`).

Other remedies. We test only one remedy, inverse-frequency class weighting. We do not test the inflection weighting of the VLN-CE baseline [Wijmans et al., 2019], which upweights changes of action, a dedicated stopping head [Xiang et al., 2020], or data rebalancing [Cheng et al., 2025].

Statistical power. Under the preregistered design, a checkpoint with a true success rate of 10% would have been confirmed with probability about 0.52, and one at 15% with probability about 0.91 (with the other candidates at 2% and every comparator at 0). The sweep therefore cannot rule out modest gains over the selected checkpoint. Each trained checkpoint's $0/61$ is consistent with a success rate of up to 10.3% when all 37 checkpoints are bounded together.

No working policy to compare. Every trained policy here fails in the simulator, so the per-class checks are observed only on failing policies. How often they would flag a policy that navigates, and whether passing them predicts success, is untested.

Three seeds and one configuration. Each treatment has three paired seeds, with one learning rate, training length, model, history length and data split. Three seeds are below the count at which interval estimates over runs are certified [Agarwal et al., 2021], so we report per-seed values beside every aggregate and run no significance test across seeds.

Causal mechanism. Action frequency, offline per-class recall and the response to weighting are consistent with one another. Frequency, however, also varies with what each action means and with the states in which it occurs, so the results show what this procedure selected without establishing why.

7 Conclusion

We studied how checkpoints are selected when a VLM is fine-tuned to navigate, a task whose success depends on stopping at the goal. In all three unweighted runs, the teacher-forced validation loss selects the same checkpoint, and that checkpoint never predicts STOP. In the simulator, two of those checkpoints match the always-forward baseline in every episode, the third differs in one, and none stops. The averaged loss gives no warning, because STOP is only 3 of the 256 decisions it averages. Balanced STOP recall ($0/64$ without weighting) and the existence gate do warn. Executing all 37 distinct saved checkpoints yields a single success, by the untrained model in one episode, and no checkpoint was shown to navigate better than the selected one. We recommend reporting recall on the action that defines success alongside any validation loss used to select checkpoints.

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A Exploratory probes in full

We ran three single-seed follow-up runs, each aimed at another possible cause of the main result: an objective that rewards constant behavior, too little data, and a model without spatial grounding. Section 5 summarizes them, and this appendix gives the protocol and the confounds. These probes are exploratory. They were planned after the main results, each has one seed, and all three share the loader confound described under Probe 1. Each was verified by independent reconstruction from raw per-decision or per-episode artifacts, and every number below is generated from those artifacts.

Probe 1: supervise four-action chunks. Under single-action supervision, always predicting MOVE_FORWARD matches 64.1% of released training decisions. Re-rendering supervision as four-action chunks cuts the best constant strategy to 25.2% of exact chunks. The loss, however, is a mean over chunk positions, and a constant MOVE_FORWARD still matches 68.0% of the positions this probe trained on. The backbone, adapter configuration, optimizer settings, observation cache, partitions and seed match the main study, but the budget is larger. This probe ran 3,941 updates, about 21.9 epochs against the main study's single epoch. The rolled-out checkpoint is update 450, selected on the same frozen selection set. Because the long run kept only its final state, that checkpoint comes from a re-run stopped at 450 updates (selection CE 0.6793). The probe also supervises four token positions per decision where the original supervised one. Because a chunk needs four future actions, the loader drops every cached decision in a trajectory's last three steps, including every decision whose target is STOP. STOP is therefore supervised only as a chunk's fourth action, and the selection set keeps 240 of its 256 decisions. Across the chunk runs, STOP is 0.47–0.57% of supervised positions, against 2.31% of the main-study training decisions, and 4 of the 960 positions on the selection set. No chunk run is trained to emit STOP as its next action, so this pipeline largely produces policies that never stop. In the simulator, the policy uses all three movement actions, and constant-action episodes fall from $61/61$ to $39/61$. It emits 0 STOP actions in 30,500 executed actions. The probe changes the objective, the supervision density and the budget together, so it does not isolate any one of them.

Probe 2: scale the data 44×. We expanded the frozen 3,072-decision training set to 40,000 and then 134,754 decisions. The sets are *nested*: each contains the frozen list as a verified position-for-position prefix, drawn from the same 2,451 trajectories under the same eligibility rules and seed, so the contrast isolates the number of unique decisions. The update budget is held at 3,000, so the number of epochs falls. After the chunk filter, the two runs train on $37{,}627$ and $127{,}661$ samples, and each run's $48{,}000$ sample draws come to 1.28 and 0.38 epochs. The best selection cross-entropy is 0.6843 for the chunk run, 0.6070 at 40,000 decisions and 0.5955 at 134,754. The chunk run's value comes from the longer schedule above and is not budget-matched to the two scaled runs, and each value is one seed, so we do not rank the three. Held-out accuracy reaches 0.760 against an always-forward reference of 0.690 on the same set. Across all 20 and 20 calibration points (checkpoints scored on the selection set during training) of the two scaled runs, STOP was predicted 0 and 0 times. The chunk run's 99 scattered STOP predictions, three of them correct, occur only after two epochs, in an overfitting regime that the scaled runs never enter, so we make no causal claim about scale removing them. In the simulator, the largest run's checkpoint enters the goal radius on $13/61$ episodes (Figure 5) and emits 0 STOP actions in 30,500 actions.

Probe 3: inject explicit metric geometry. We attached a geometry branch to the frozen backbone at the largest budget. The simulator's ground-truth metric depth for each frame is backprojected

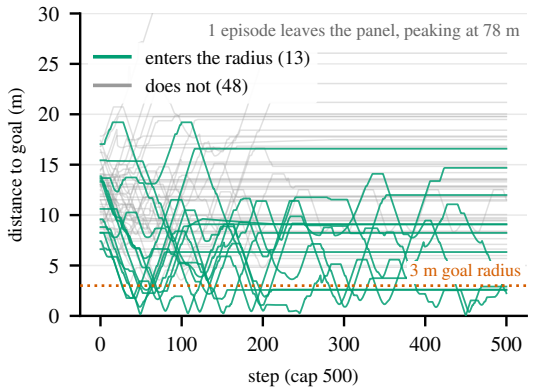


Figure 5: **The largest probe run in closed loop.** Distance to goal over the 500-step cap for the 61 episodes of the exploratory 134,754-decision run of Probe 2, whose chunk loader (Probe 1) dropped every decision whose target was STOP. Green enters the 3 m radius, grey does not. Thirteen of the 61 enter it, against the $1/61$ Oracle Success every main-study checkpoint records in Table 2. No episode ends with STOP.

through the camera model and aggregated to 64 patch points per frame, with invalid pixels masked. An equivariant geometric-algebra transformer processes these points through a readout that preserves metric scale and egocentric direction, and its output enters the model as 256 additional tokens whose attention phases carry a multi-scale 3D positional code (wavelengths 0.60–12 m). Replacing real depth with a flat wall shifts the branch's output by 0.63 of its own standard deviation at the selected checkpoint, so the branch carries information about the scene. The model also uses it: masking the tokens at fixed weights moves selection cross-entropy from 0.6407 to 1.0625. Trained on the same data, adapter, seed, budget and selection set as the geometry-free run, plus the branch's own 17.6M parameters at learning rate 10^{-4}, the geometry arm reached 0.6407 against the geometry-free baseline's 0.5955 at the same budget. Both values are single-seed and fall on the last scored checkpoint, and seed variation alone can separate learning curves that share a configuration [Henderson et al., 2018]. The geometry arm predicted STOP 0 times across 20 calibration points, and in the simulator it entered the goal radius on $9/61$ episodes while emitting 0 STOP actions in 30,500 actions. Published results on depth or 3D input to VLM navigation policies point in both directions. Adding simulator depth to a fine-tuned VLM lowered R2R-CE `val_unseen` success from 27.1 to 23.1, which the authors attribute to the model's limited ability to process depth [Zhang et al., 2025]. Entity-level 3D soft tokens injected into a frozen VLM improved object-goal navigation [Wu et al., 2026]. We know of no measurement of the configuration used here, per-patch metric-depth geometry tokens with a multi-scale 3D positional code on an instruction-following R2R-CE policy. This single-seed result adds one data point to a mixed record.

Summary and limits. Each probe changed one mechanism that could explain the constant-action behavior (the objective's payoff, the amount of data, the lack of spatial structure), and behavior changed. Constant-action episodes fell, the largest run entered the goal radius far more often, and the model used the geometry tokens. These results do not bear on stopping. Summed over every calibration point of the scaled and geometry runs, STOP was predicted 0 times at 13× data, 0 at 44×, and 0 with geometry, but the loader described under Probe 1 removed every STOP-target decision from training and selection. These zeros, like the absence of STOP in the simulator, are therefore largely a product of the pipeline, and the probes cannot show that the selection loss's blindness to STOP persists under these changes. Each probe is one seed at one configuration and was planned after the main results. The 134k and geometry runs reach their lowest selection cross-entropy at their final calibration point (the 40k run's minimum is interior, at update 2700 of 3,000), so no convergence claim is available for the two largest runs. The geometry probe adapts the frozen backbone only through a rank-32 attention-only adapter, a configuration no VLN-CE system we know of reports using, and it reads the simulator's ground-truth depth, so it bounds only this configuration.

B Related work in full

Offline metrics versus closed-loop competence. In neighboring embodied domains, it is established that offline imitation metrics can fail to identify good closed-loop policies. In driving, offline prediction error is not necessarily correlated with driving quality, and two models with identical prediction error can differ widely when deployed [Codevilla et al., 2018]. Open-loop metrics on a modern planning benchmark favor degenerate planners that nearly ignore their input [Dauner et al., 2023]. In manipulation, the robomimic study reports that the best validation loss does not correspond to the best-performing policy, with validation-selected checkpoints 10–100% worse than rollout-selected ones [Mandlekar et al., 2021]. Hao et al. [2023] report that choosing a trained policy's input-modality mask by lowest validation loss picked a zero-success mask in all three seeds on a robomimic task, while choosing among separately trained policies by the same loss roughly doubled another baseline's success. Simchowitz et al. [2025] construct a setting in which rollout cost rises while validation loss falls throughout training. By 2026, a weak correlation between action-space validation error and success is described as a practitioner consensus [Huang et al., 2026], although the same study finds raw validation error rank-consistent with rollout success when only the number of training steps varies. Behavior-cloned policies can reach near-zero held-out error while relying on spurious correlates that collapse under their own deployment distribution [de Haan et al., 2019], including collapse toward a dominant action regime overrepresented in demonstrations [Codevilla et al., 2019]. Codevilla et al. [2018] also show that the offline–online correlation improves substantially with better-chosen validation data and metrics. Our claim therefore concerns only the plain cross-entropy over decisions in their natural action frequencies, which general-purpose fine-tuning stacks compute when rollouts are unavailable. The divergence also has limits. Under log-loss with a discrete action vocabulary, compounding error is at most polynomial in the horizon [Foster et al., 2024], and the exponential-in-horizon bound of Simchowitz et al. [2025] applies to continuous actions only. The polynomial bound does not rule out compounding error, or the shift from expert to policy histories at deployment, and our experiments do not separate their contribution from action imbalance. The mechanism we point to, that a frequency-dominated cross-entropy is blind to a rare class, was stated for STOP in the training loss by Xiang et al. [2020] and for rare action changes in validation accuracy by Wijmans et al. [2019], who also chose among five low-loss checkpoints by executing them and found inflection-weighted cross-entropy a better predictor of end-to-end performance than plain cross-entropy. None of this prior work on offline metrics combines instruction-following navigation, a VLM policy and a success criterion that depends on a single rare action, which is the combination studied here.

The stopping problem in VLN. Stopping failure is also documented within VLN, on the discrete navigation graph. Zhao et al. [2023] quantify a persistent gap between Oracle Success and Success (agents pass the goal and fail to stop) across four systems. Two further findings concern learned action regularities. A case study of one agent finds its errors reflect a systematic bias toward forward actions learned during training [Yang et al., 2023], and Thomason et al. [2019] note that peaked action distributions in R2R training let agents memorize simple rules, such as moving forward an average number of steps. Xiang et al. [2020] treat stopping as a distinct learning problem and add a dedicated head. For a VLM fine-tuned for R2R-CE, Dang et al. [2026] report Oracle Success of 71.6% against success of 58.6% on `val_unseen` and conclude that the model "lacks the precision for the terminal stopping maneuver". They evaluate a trained model and do not study how its checkpoint was chosen. Fine-tuning a small VLM for aerial navigation, Tamazyan et al. [2025] find performance "critically affected by a severe action imbalance in the training data". In VLM fine-tuning for R2R-CE, Cheng et al. [2025] name the underrepresentation of the stop action as a label-imbalance problem and report that removing their data rebalancing lowers `val_unseen` success from 49.7 to 30.0. Their remedy acts on the data blend. The failure reported here, never stopping at the goal, survives inverse-frequency *loss* weighting (the weighted checkpoints press STOP seven times, all outside the goal radius), and the object of our study is the selection metric. Poor stopping is already documented, and we add a closed-loop measurement, in continuous R2R-CE with a fine-tuned VLM and every saved checkpoint executed, of what an averaged selection loss reports while the policies it ranks lose the terminal action. The loss fell from 5.49 nats to below 1 in every run, and no checkpoint was shown to navigate better than the one it chose. Every trained unweighted checkpoint predicts STOP on none of the 256 selection decisions (update 0 predicts it 28 times), so in those runs any rule choosing among trained checkpoints picks one that never predicts STOP there. The collapse is a training outcome, and the averaged loss did not flag it.

VLM policies for continuous navigation. Recent systems fine-tune large vision–language models to act directly in continuous environments, building on Room-to-Room (R2R) and R2R-CE [Anderson et al., 2018, Krantz et al., 2020, Hong et al., 2022]. These systems differ widely in action interface, observation history and planning scaffolding, so we make no comparison to published success rates and report only within-protocol contrasts against references frozen in advance. R2R-CE papers that state a selection rule select on closed-loop `val_unseen` performance, and VLN-CE performs early stopping on it [Krantz et al., 2020]. NaVILA, a VLM policy, trains for one epoch and states no selection rule [Cheng et al., 2025]. We found no R2R-CE paper that selects by teacher-forced loss. We study that rule because it is what a general-purpose fine-tuning stack computes when rollouts are unavailable. The Hugging Face Trainer, asked to keep its best checkpoint, keeps the one with the lowest evaluation loss unless given another metric [Hugging Face, 2026]. Because `val_unseen` stays sealed, the closed-loop evaluation on reserved trajectories is a diagnostic of the rule, and we make no leaderboard claim. Class-imbalance remedies such as re-weighting are standard [Cui et al., 2019], and our second treatment arm applies one and measures what it changes. The VLN-CE baseline itself trains with *inflection weighting* [Wijmans et al., 2019], which upweights timesteps where the action changes. The two remedies are not interchangeable. On our 3,072-decision training set every STOP is an inflection, so inflection weighting would give it $2.48\times$ the weight of a non-inflection step ($2.55\times$ over the released split, and the VLN-CE baseline configures 3.2 for R2R [Krantz et al., 2020]), the same factor as any turn onset. The inverse-frequency weighting we test gives STOP $28.5\times$ the weight of MOVE_FORWARD on that same set. The two rules weight different groups of decisions, rare classes against changes of action, so these factors do not rank the remedies' strength. We did not test inflection weighting. The failure to stop at the goal survives the weighting we did test.

C Exact ties in the action scorer

Every predicted action reported here is the action token with the lowest negative log-likelihood. The logits are in bfloat16 (BF16), and in every stored decision the two best scores are either equal or at least 0.125 nats apart; an exact tie goes to the lowest official ID, so STOP wins every tie it is in and MOVE_FORWARD wins every tie with a turn. Recounted from the four scores stored for each decision, ties account for 2 of the 28 STOP predictions at update 0; for 15 of the 34 and 3 of the 23 at the weighted spikes of Figure 4; for 2 of seed 20260827's 6 balanced STOP hits and none of the other weighted seeds' 12 and 1; and for 2 of the 7 closed-loop STOP presses, one in each seed that stopped, both on step 1. Strict preferences alone give 26, 19 and 20 predictions, balanced STOP recall of $12/64$, $1/64$ and $4/64$, and 5 presses; weighted balanced accuracy falls from a mean of 0.397 to 0.375. Both recall and accuracy still exceed the unweighted run in every pair. Many strict wins are narrow: 11 more of the 34 at the seed-20260826 spike win by a single BF16 step, and all seven presses won by at most 0.5 nats, with restricted STOP probability 0.28–0.40. No zero STOP count in this paper can hide a tie, because a tie that includes STOP is scored as STOP, and at every in-radius decision STOP trailed the chosen action by at least 0.75 nats. Unweighted seeds 20260825 and 20260826 have no tie in their $61{,}000$ closed-loop decisions; seed 20260827 has five, each a step-1 tie between MOVE_FORWARD and a turn in an episode that then matches always-forward. The chunk run's 99 STOP guesses (Appendix A) cannot be split this way, because that run stored only counts.

D Reproduction details

Backbone and adapter. `Qwen/Qwen3-VL-8B-Instruct` at Hugging Face revision `0c351dd01ed87e9c1b53cbc748cba10e6187ff3b`, run through its own processor and visual pathway. The LoRA adapter has rank 32, $\alpha = 64$ and dropout 0, on the query, key, value and output projections of each of the 36 language-model attention blocks, for $30{,}670{,}848$ trainable parameters. Every other weight stays frozen in BF16. The adapter and optimizer state use 32-bit floats.

Decision sampling. Each physical trajectory keeps one of its instructions, chosen by a seeded hash. A decision at step t is eligible when $3 \le t \le 96$, so no training or held-out history needs padding. The selection, natural-outcome and balanced partitions take one decision per trajectory by seeded hash, the balanced one drawing 64 per action from distinct trajectories. Training takes one decision

Table 3: Non-learning evaluator controls on the 61 frozen `val_seen` development episodes. Stochastic rows report means over five fixed seeds.

Control	Runs	Episodes/run	SR (%)	SPL (%)	nDTW
Released trajectory replay	1	61	100.00	96.79	1.000
Immediate STOP	1	61	0.00	0.00	0.204
Always forward	1	61	0.00	0.00	0.234
Invalid action	1	61	0.00	0.00	0.204
Official random	5 seeds	61	0.98	0.79	0.233
Official handcrafted	5 seeds	61	0.98	0.95	0.245

from each of its 2,451 trajectories and 621 more from the same trajectories, again by seeded hash, and visits them once in a seeded order.

Class weighting and selection. The weighted arm computes one weight per action from the 3,072 training decisions, $w_a = N/(4n_a)$, so that the mean weight over those decisions is one: STOP 10.82 ($n = 71$), MOVE_FORWARD 0.379 (2,024), TURN_LEFT 1.430 (537) and TURN_RIGHT 1.745 (440). Each decision contributes $w_y(-\log p_y)$ for its expert action y. The 16 decisions of an update are each divided by 16 and accumulated, with no renormalization of the weights within an update. The weights act on training only: checkpoint selection and every cross-entropy reported here use the unweighted four-way loss in both arms.

Prompt, history and action tokens. A system message lists the four action IDs and asks for an answer of the form "Action ID: N". The user message holds the four frames in time order, each preceded by a label such as "Frame t-1 (previous action: 1=MOVE_FORWARD).", and ends with the instruction and "Choose the next official action now." The target "Action ID: N" tokenizes to four shared tokens (ids 2512, 3034, 25, 220) and one digit token: 15 for STOP ($N = 0$), 16 for MOVE_FORWARD ($N = 1$), 17 for TURN_LEFT ($N = 2$) and 18 for TURN_RIGHT ($N = 3$). Only the digit position is supervised, and every score compares the four digit tokens at that one position. In closed loop, a frame that does not yet exist is replaced by a copy of the initial observation and labeled as such, with no previous action. Training never sees such a padded history, and six of the seven weighted STOP presses fall on step 1, where it occurs, so those early stops are confounded with this untrained input.

Runtime. Training ran on one NVIDIA GB10 GPU (aarch64) and the first closed-loop evaluation on the same aarch64 platform, with Habitat-Sim 0.3.3. All 37 policies were later executed and scored on NVIDIA L40S GPUs (x86-64, Appendix F). Because some stored decisions are exact BF16 ties (Appendix C), another GPU model could change individual actions.

Protocol records. Each training run records the SHA-256 hash of every protocol file it followed. The protocol for the two later seeds was edited after their runs finished, to mark it completed and add a closing note. No rule, seed or threshold changed, and those four runs record the hash of the version they followed.

E Closed-loop failure structure

Figure 6 plots the closed-loop behavior summarized in Table 2 and the final-error contrasts below.

Lower final error does not mean better navigation. Weighting reduces mean final navigation error by 0.677 m, with descriptive scene-cluster interval $[-1.615, +0.010]$. However, it worsens mean closest approach by 0.406 m, interval $[+0.064, +0.754]$, and changes nDTW by -0.005, interval $[-0.031, +0.018]$. Weighted rollouts leave the agent stationary (executing a turn or a collision-blocked forward, with no position change) on 94.9–96.1% of steps and visit a median of 1, 2, and 1 unique positions, versus 18 unweighted. Lower final error and collisions mainly reflect stationary turns and premature STOP actions that replace forward moves, while the closest approach to the goal gets worse.

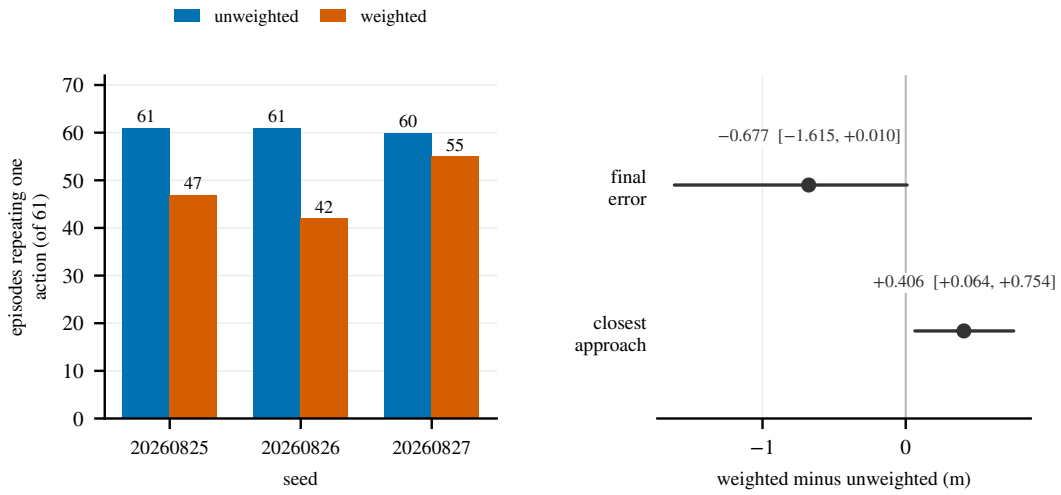


Figure 6: **Closed-loop failure structure.** Left: most weighted episodes repeat one action for the full rollout. Right: weighting lowers final navigation error but worsens closest approach (positive minimum-distance delta); lines show descriptive 95% intervals from resampling the 53 scenes. SR, OSR, SPL, and SDTW do not improve.

Table 4: Every saved checkpoint executed on the 61 main episodes on NVIDIA L40S GPUs: STOP presses, with oracle successes in parentheses. Bold: the selection rule's pick, re-run here. Every trained checkpoint succeeds in $0/61$ episodes: exact two-sided 95% interval 0–5.9% for one checkpoint alone, and at most 10.3% when all 37 are bounded together (one-sided, Bonferroni). Update 0, one untrained model shared by all six runs, succeeds in $1/61$ (bounded together: at most 13.7%), enters the radius in 1 and presses STOP 50 times.

		Update					
Treatment	Seed	32	64	96	128	160	192
Unweighted	20260825	0 (1)	3 (1)	0 (1)	0 (1)	**0 (1)**	0 (1)
	20260826	8 (0)	5 (1)	0 (1)	0 (1)	**0 (1)**	0 (1)
	20260827	7 (1)	0 (1)	0 (1)	0 (1)	**0 (1)**	0 (0)
Weighted	20260825	3 (0)	6 (0)	0 (1)	4 (0)	8 (1)	**5 (1)**
	20260826	22 (0)	42 (0)	0 (0)	**0 (1)**	1 (0)	6 (1)
	20260827	25 (0)	0 (0)	0 (2)	0 (1)	15 (0)	**3 (0)**

F Executing every saved checkpoint

Policies and hardware. Each run saved seven checkpoints, at updates 0, 32, ..., 192. In every update-0 file each adapter output matrix is exactly zero, so all six compute the frozen base model; the unweighted seed-20260825 copy stands for them. That leaves 37 distinct policies: the 30 trained checkpoints the rule did not select, the untrained model and the six selected checkpoints. All 37 ran on the same 61 episodes, in the same order, with the same simulator settings, step cap and tie rule as the first closed-loop evaluation, on NVIDIA L40S GPUs (driver 610.43.02), two processes per GPU, in an Apptainer image of NVIDIA's PyTorch 25.10 container (torch 2.9.0a0, transformers 4.57.6, peft 0.18.1, habitat-sim 0.3.3). The preregistration was frozen in version control on 9 October 2026, before any unselected checkpoint was executed or scored. It was not filed with a public registry, and the frozen text is in the code release. Every result records the frozen version, the GPU's name and the container image's hash. Table 4 gives every policy's STOP presses and oracle successes. All 2,257 episode runs completed, and none needed a retry after an error.

Pre-run check. Before the preregistration was frozen, the weighted seed-20260825 checkpoint at update 192 was scored offline and executed on eight `val_seen` episodes in neither suite (episodes 4, 5, 10, 11, 13, 14, 16 and 17): alone, two at once on one GPU, and across a forced stop and resume.

The same job replayed the expert and an always-forward policy on all 61 main and 61 confirmation episodes. Expert replay ended within 3.2×10^{-6} m of the first platform's final distance on every main episode, the offline scores reproduced the training records within the preregistered bars, and two processes kept the GPU 94% busy. An earlier run also passed; the freeze relies on a repeat after later code changes.

Offline scores. As the first job after the freeze, all 37 policies were scored on the 256 selection, 640 natural and 256 balanced decisions, before any rollout was read. Of the $14{,}848$ decisions training had also recorded, 90 are bit-identical; the largest change in a mean cross-entropy is 0.043 nats, within the preregistered 0.05, and no decision won by a nat or more changed its action.

The first comparison: a better checkpoint. For each of the 31 candidates the null hypothesis is that it succeeds on no more episodes than its comparator: its own run's selected checkpoint, and for the untrained model each of the six selected checkpoints, all of which it must beat. With b the episodes only the candidate solves and c those only its comparator solves, $p = P(\mathrm{Binomial}(b + c, 1/2) \geq b)$, rejected under the alpha-consistent form of Tarone's screen [Tarone, 1990, Hommel and Krummenauer, 1998]: with every comparator at zero successes, a candidate needs at least 5 of 61 when it is the only one that could reach significance. Every trained candidate has $b = c = 0$, so $p = 1$; the untrained model has $b = 1$, $c = 0$ against each selected checkpoint, so $p = 0.5$. No candidate clears the bar, so none goes on to a confirmation run.

The second comparison: choosing by balanced loss. This rule takes, within each run, the checkpoint with the lowest balanced-set cross-entropy, ties to the earlier update. It was named before any unselected checkpoint was executed or scored, though after the selected checkpoints' balanced scores were known. The balanced-loss rule picks updates 64, 192 and 160 in the unweighted runs and 160, 192 and 160 in the weighted runs, the selection rule's own checkpoint in one run of six. Its picks solve no episode, nor do the selection rule's, so $b = c = 0$ and $p = 1$: the test fails, and the balanced loss is reported as a warning, not as a way to choose. Keeping each run's last checkpoint solves none either, so it did as well as the balanced-loss rule, and class balance is not shown to matter here. Two more rules are reported only as description, with unadjusted p-values. The lowest selection loss among checkpoints that pass the existence gate abstains in all six runs, because none passes. The highest strict balanced STOP recall, ties to the lower balanced loss, picks the untrained model in the three unweighted runs and so solves one episode, 358, where the selection rule solves none ($p = 0.5$, unadjusted, descriptive); this is not a test, and we do not call that rule better.

Selection regret. In each run, the most successes among its seven checkpoints minus the selected checkpoint's is 1, in all six runs, from the untrained model's episode 358. This is biased upward: it is the largest of seven counts on the same 61 episodes.

The offline STOP count. All 18 trained unweighted checkpoints predict STOP on none of the 256 selection decisions. The three selected ones pressed it in no episode on the GB10, and again in none here. We predicted that the 15 unselected ones would press it in no episode either. The prediction failed at update 64 of seed 20260825 (3 presses), updates 32 and 64 of seed 20260826 (8 and 5) and update 32 of seed 20260827 (7): 23 presses, all at the first step, 6.6–12.9 m from the goal.

Offline criteria against outcomes. No criterion can be associated with success, which only one policy reached. Within runs, checkpoints with a higher selection cross-entropy pressed STOP more often (mean within-run Spearman 0.56, permutation $p = 0.0011$, below the preregistered 0.0025); the association with oracle success does not reach that threshold ($p = 0.0027$). The existence gate failed at all 37 policies; balanced STOP recall was at least 16 of 64 at three of them, none of which succeeded.

The two platforms. The six selected checkpoints ran on the GB10 first and on the L40S here. Of their 366 episodes, 347 have identical action sequences and 364 identical success, oracle-success and stop flags; each of the other 19 first differs at a decision the GB10 won by 0.125 nats or less, so the two machines agree up to near-ties. The two flag differences are in one weighted checkpoint: one episode ends in STOP on the L40S only, and another enters the goal radius on the GB10 only; neither succeeds on either. The comparison cannot say why two rollouts differ: the GPU, its kernels, the renderer and the runner code all differ at once.